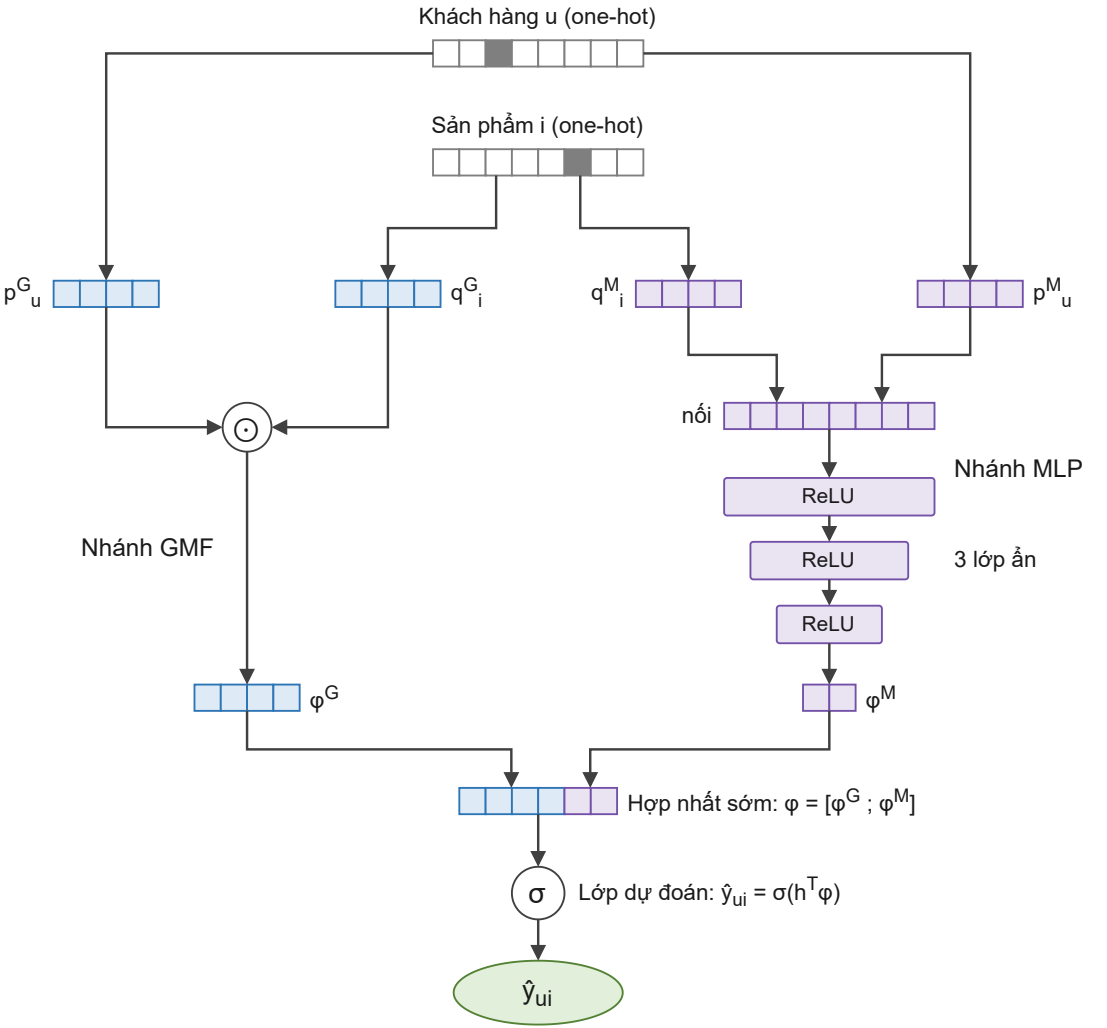
Khách hàng u (one-hot)
Sản phẩm i (one-hot)
p^G_u
q^G_i
q^M_i
p^M_u
nối
Nhánh MLP
ReLU
ReLU
ReLU
3 lớp ẩn
Nhánh GMF
φ^G
φ^M
Hợp nhất sớm: $\varphi = [\varphi^G ; \varphi^M]$
σ
Lớp dự đoán: $\hat{y}_{ui} = \sigma(h^T\varphi)$
$\hat{y}_{ui}$